Predicting Content Decline for Editorial Review

A retrospective client-grouped evaluation of interpretable ranking rules and machine-learning models on March 2026 search-performance data

Yasmeen Azmat Ali · 10 October 2026 · Research lane: Refresh / Content Opportunity Scoring

Abstract

Objective. To determine whether first-half search and engagement signals can prioritize content for human editorial review before an observed second-half impressions decline. Methods. March 2026 daily FlyRank warehouse data were aggregated to 92,548 eligible client–content pairs; a seed-42 client-grouped split placed 24,886 items from 10 unseen clients in the holdout. Transparent ranking rules, a depth-3 decision tree, logistic regression and random forest were compared using ROC–AUC, average precision and Precision@500. Results. Logistic regression achieved ROC–AUC 0.749 and Precision@500 0.882; a momentum-only ranking achieved ROC–AUC 0.754 and Precision@500 0.874. Across five grouped folds, mean ROC–AUC was 0.669 for logistic regression, 0.688 for random forest and 0.670 for momentum. Interpretation. A simple momentum measure is competitive on the reported holdout, but client concentration, missing GSC observations and fold variability constrain claims. Conclusion. The evidence supports a human-reviewed prioritization workflow, not automated page changes or a validated prospective forecast.

Keywords: content decline; editorial triage; search analytics; grouped validation; target leakage; interpretable ranking

1. Introduction

Editorial teams cannot inspect every content item simultaneously. A review queue should place likely deteriorating pages near the top while minimizing wasted analyst time. False positives consume scarce review capacity; false negatives leave potentially important pages unchecked. The operational decision studied here is therefore which pages to review first, rather than whether an automated system should rewrite or remove them.

Research question. Can signals observed by 15 March 2026 rank pages at risk of an impressions decline during 16–31 March among previously unseen clients, and does model complexity materially improve the resulting review queue over a transparent momentum-only rule?

The paper consolidates the documented ML-02–ML-10 internship work. The 30,000-row starter dataset informed task framing and leakage analysis, while the main reported model comparison uses a separate 92,548-item March warehouse cohort. Their metrics and populations are not pooled.

2. Data, outcome and eligibility

The data source is the FlyRank ML Internship fact_content_daily_performance warehouse table for 1–31 March 2026. The initial audit records 9,841,378 client–content–day observations, 55 clients and 331,437 distinct content items. Eligibility requires at least 50 GSC impressions during 1–15 March, leaving 92,548 client–content pairs from 40 clients for modeling. Raw client identifiers, private queries and row-level records are omitted from this public report.

Population / window	Notebook-reported quantity
Warehouse daily observations	9,841,378
Warehouse clients / distinct content items	55 / 331,437
Eligible modeling pairs / clients	92,548 / 40
Training split	67,662 items / 30 clients
Unseen-client holdout	24,886 items / 10 clients
Observation period	1–15 March 2026
Retrospective outcome period	16–31 March 2026

The binary label equals 1 when the second-half impression count is strictly less than 80% of the first-half count. This is a count-based, retrospective proxy; the windows span 15 and 16 days, respectively, and were not normalized by observed days. Train prevalence was 25.6%, compared with 36.9% in the held-out clients.

3. Methods

3.1 Feature construction and model families

The March 15 decision cutoff governs predictors. Reported features include log-scaled first-half impressions and pageviews, first-half click-through rate (CTR), average position, within-first-half momentum, engagement rate and an indicator of GA4 data availability. Hashed client and content IDs were retained for grouping and diagnostics but excluded as ordinary predictors. Evaluated methods were a volume-only ranking, an honest first-half Week-4 rule, a depth-3 decision tree, logistic regression, random forest and a momentum-only ranking.

3.2 Evaluation design

The reported 75/25 client-grouped split used seed 42 and had zero client overlap. ROC–AUC measures discrimination across thresholds; average precision summarizes precision–recall performance; Precision@500 measures the share of truly declining items among the first 500 review candidates. Five additional client-grouped folds provide a sensitivity check on split dependence. The fixed 500-item budget is a ranking evaluation choice, not an operationally verified staffing capacity.

3.3 Leakage and label integrity

An earlier starter-data audit exactly reconstructed all 30,000 labels from impression-window inputs and found perfect agreement between the label and trend_direction; these fields cannot be treated as independent predictors of that same target. In the warehouse leakage experiment, adding future-period impressions inflated performance, and a deliberately leaky Week-4 reference later yielded AUC 0.911. That reference is excluded from fair comparisons. Explicit second-half variables were removed from the final predictor matrix, although exact warehouse reporting timestamps and late corrections remain unresolved.

4. Results

4.1 Primary held-out comparison

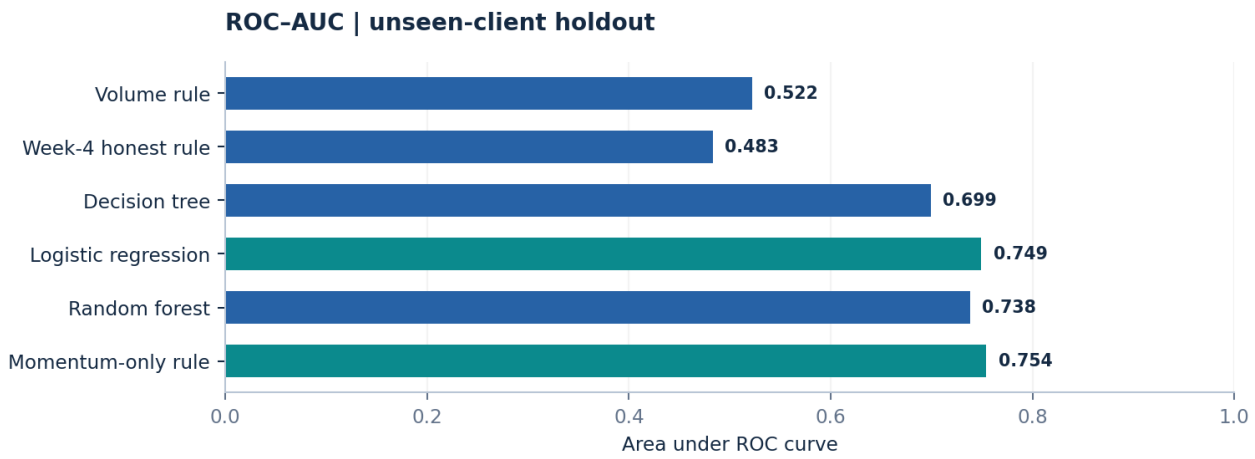


Figure 1. ROC–AUC on 24,886 content items from 10 held-out clients. All six legitimate methods use the reported grouped holdout.

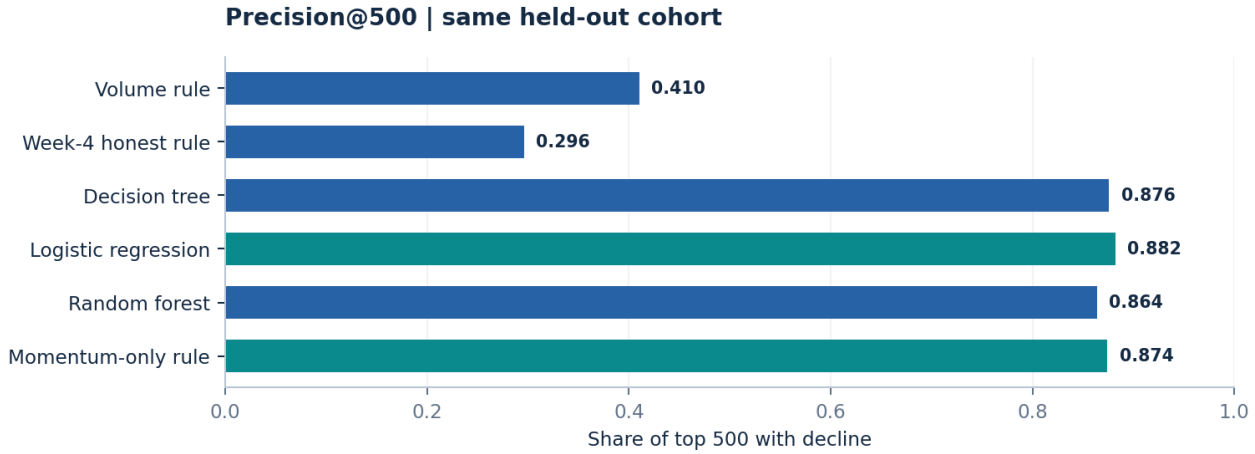


Figure 2. Precision among the top 500 review candidates on the same held-out cohort. Values are descriptive of one split.

Method	ROC–AUC	Average precision	Precision@500
Volume-only rule	0.522	0.381	0.410
Week-4 honest rule	0.483	0.361	0.296
Decision tree (depth 3)	0.699	0.563	0.876
Logistic regression	0.749	0.644	0.882
Random forest	0.738	0.632	0.864
Momentum-only ranking	0.754	Not reported	0.874

Table 1. Notebook-reported held-out model metrics. “Not reported” is not interpreted as zero. The deliberately contaminated reference is omitted.

Logistic regression placed approximately 441 declining items among its top 500, compared with 437 for momentum alone: four additional items, or 0.8 percentage points, on this holdout. Momentum-only ranked marginally higher by ROC–AUC (0.754 versus 0.749). These small, metric-dependent differences do not establish robust superiority.

4.2 Grouped-fold stability and feature interpretation

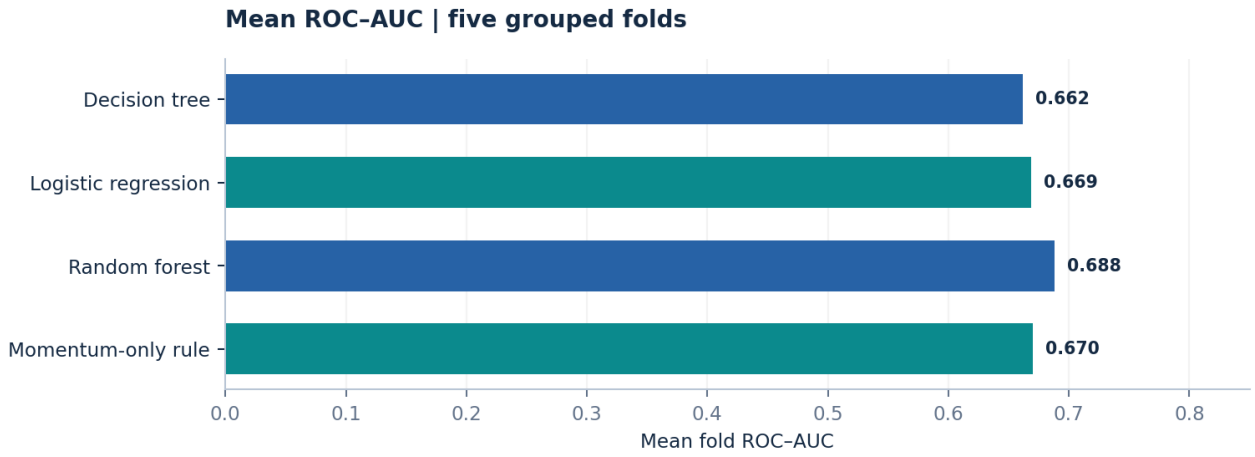


Figure 3. Mean ROC–AUC across five client-grouped folds. These are fold means, not confidence intervals; reported fold ranges were tree 0.603–0.707, logistic 0.602–0.754, forest 0.658–0.736 and momentum 0.629–0.747.

The random forest had the highest reported mean grouped-fold AUC (0.688), while the momentum-only rule led on the single designated holdout. The overlapping fold ranges and changes in rank order argue against a strong claim that one model family is consistently best.

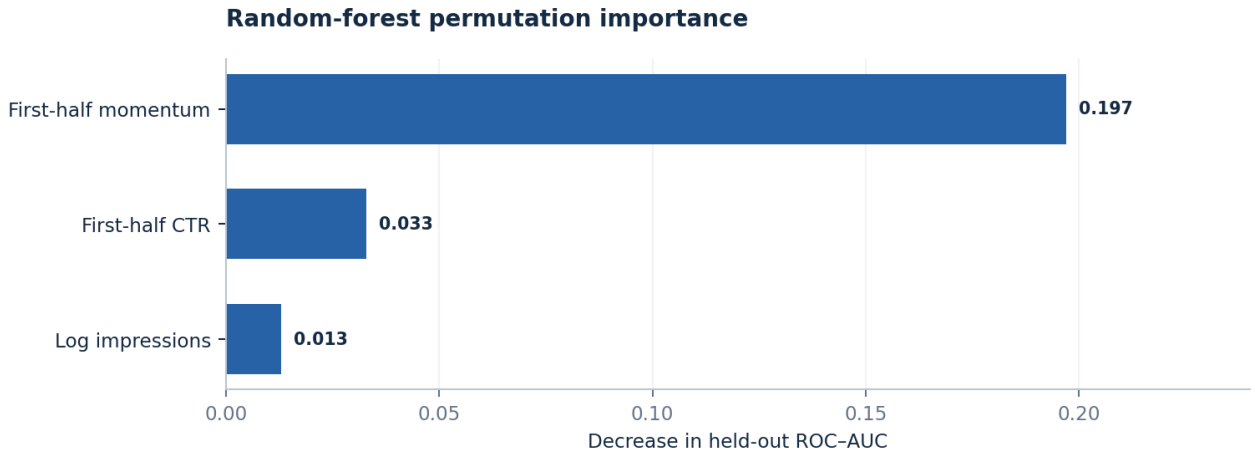


Figure 4. Selected random-forest permutation importance values on the held-out clients. Values reflect the decrease in ROC–AUC after shuffling a feature; they do not measure causal influence.

First-half momentum dominated the reported forest permutation audit (approximately 0.197 AUC decrease), far ahead of CTR (0.033) and log impressions (0.013). At a fixed 0.5 forest classification threshold, the notebook reported 14,950 true negatives, 764 false positives, 6,838 false negatives and 2,334 true positives. These threshold errors reinforce the choice to evaluate ranked review rather than fully automated binary decisions.

4.3 Supporting signal audit

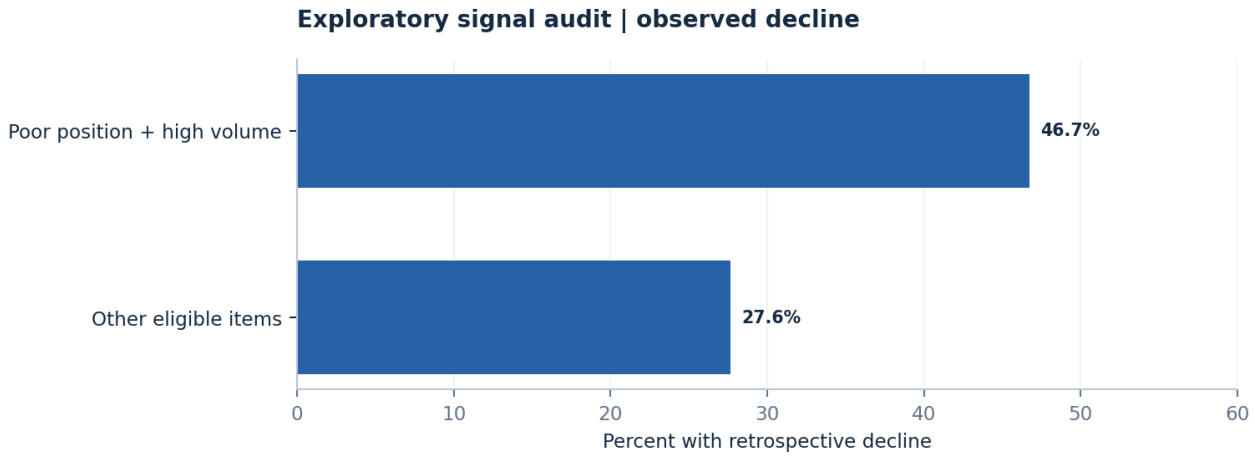


Figure 5. Retrospective decline rate for the ML-06 poor-position/high-volume flag (n=4,910) versus unflagged items (n=87,638). The 19.09-point gap is an association, not an estimated intervention effect.

The flag combined first-half average position above 20 with first-half impressions at or above the 75th percentile. The flagged group had 46.72% observed decline versus 27.63% in the unflagged group. This association suggests a review signal worth investigating, but it does not demonstrate that editing flagged pages will recover impressions.

5. Measurement quality and sensitivity

First-half GA4 history was present for 32,976 of 92,548 eligible pairs. In the future outcome window, 12,198 eligible items lacked complete second-half GSC coverage. Zero-filled source values may conflate missing observations with genuine zero traffic; structural checks alone cannot resolve this measurement ambiguity.

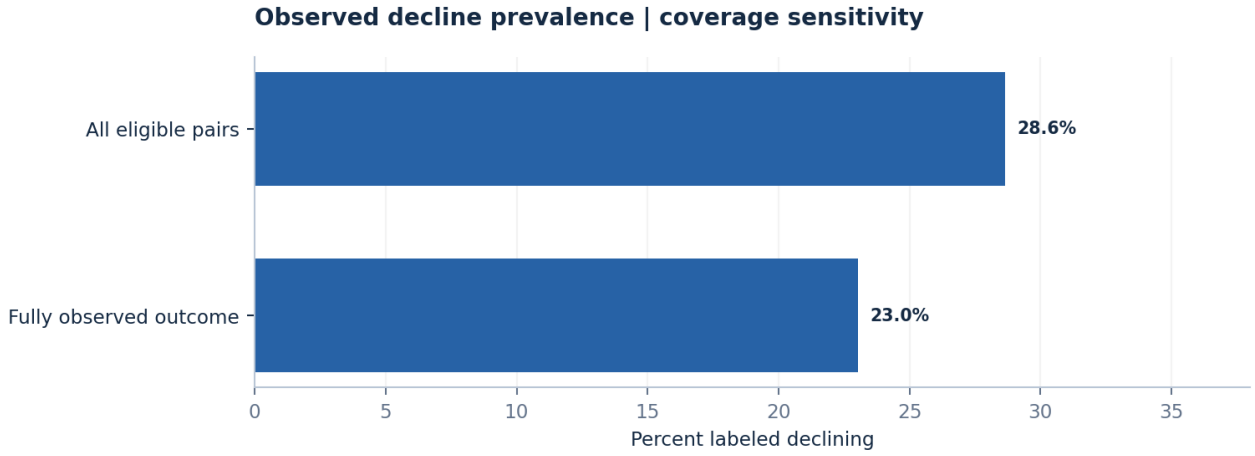


Figure 6. Retrospective label prevalence for the full eligible population (28.64%) and fully observed outcome subset (23.04%). Denominators differ, so this is a sensitivity comparison, not a causal effect.

A separate high-coverage model sensitivity subset retained 81,665 items; on its 20,306-item held-out subset, forest AUC was 0.707 and momentum-only AUC 0.716. These values are not directly comparable to the full-cohort holdout as if they were an intervention on missingness. In addition, 86.4% of the forest top 500 came from the three largest held-out clients, raising a fairness and workload-allocation question for cross-client review queues.

6. Discussion and key findings

Finding 1 — Complexity is not an automatic advantage. The momentum-only rule matched or slightly exceeded the trained models on held-out ROC–AUC, and lagged logistic regression by only four positives at a 500-item review budget. Its transparency makes it a serious comparator for a practical triage system.

Finding 2 — Validation design changes the story. The single holdout favors momentum by AUC, while five grouped-fold means favor the forest. No method exhibits demonstrated dominance across the reported checks; future-month and multi-client evaluations are necessary.

Finding 3 — Data availability is part of the prediction problem. Missing GSC observations affect the retrospective outcome label; sparse GA4 availability constrains engagement features. The model can be structurally leakage-free yet still predict an unstable or incompletely measured target.

Finding 4 — Ranking quality is not editorial uplift. Neither AUC nor Precision@500 estimates the causal effect of refreshing a page. An actionable system needs human verification, reason codes, a defined review budget and subsequent intervention evaluation.

7. Editorial action playbook

ML-10 exported a separate operational review queue spanning 331,437 content items across 55 clients. It is broader than the 92,548-item modeling cohort and must not be described as independently evaluated over its full extent.

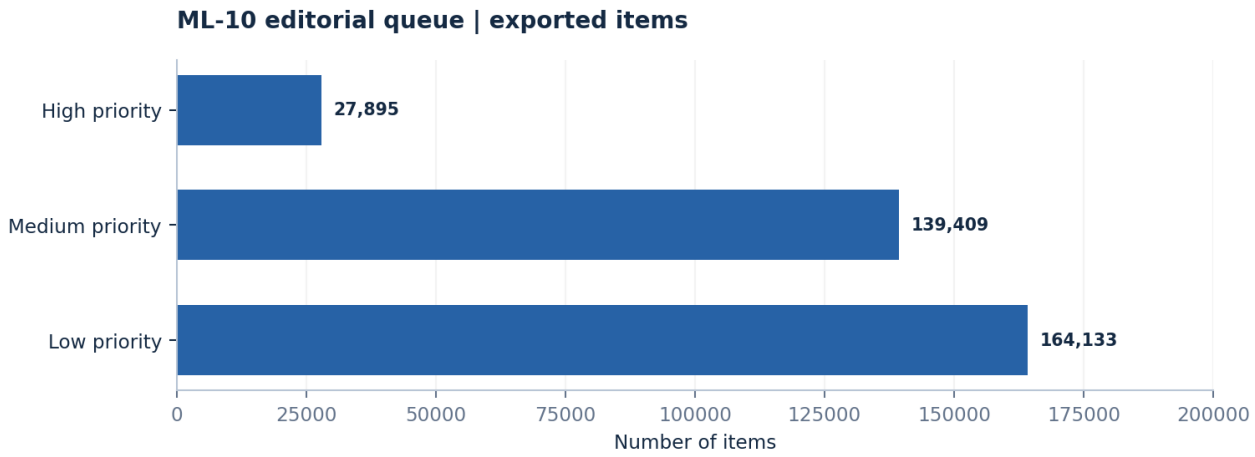


Figure 7. ML-10 exported review tiers: high 27,895; medium 139,409; low 164,133. Counts describe queue composition, not measured intervention outcomes.

Priority 1 — Evidence-first human triage. Review the highest-ranked items with GSC trends, query intent and content context visible; do not trigger automated publishing actions. Priority 2 — Reason-specific checks. For HIGH_TRAFFIC_LOW_CTR, inspect title/snippet relevance and intent; for VERY_HIGH_RELATIVE_TRAFFIC, protect strong pages from unnecessary changes; for VERY_LOW_RELATIVE_CTR, check measurement and search presentation. Priority 3 — Capacity-aware deferral. Hold GENERAL_REVIEW items for remaining analyst capacity and report the queue’s client mix.

Proposed monitoring investigation thresholds from the action playbook are ROC–AUC below 0.70, average precision below 0.58 and Precision@500 below 0.80, alongside material shifts in input coverage and client composition. These are proposed governance triggers; no live monitoring deployment is established by the uploaded reports.

8. Limitations and future work

This is a retrospective, observational, single-month study. The label is a threshold-defined proxy; the two periods are of unequal duration; incomplete GSC coverage may change labels; and client grouping does not substitute for forward-in-time validation. Differences in client base rates and client size affect both discrimination and the composition of top-k recommendations. The uploaded evidence does not establish a randomized intervention, external generalization, causal traffic lift, probability calibration or verified deployment readiness.

A stronger next evaluation would (i) test later months prospectively; (ii) define day-normalized or observed-day-aware outcomes; (iii) audit warehouse availability timestamps; (iv) report per-client Precision@k and workload concentration; (v) compare transparent and trained rankings on identical future cohorts; and (vi) measure editorial review outcomes and intervention effects separately from prediction accuracy.

9. Reproducibility and evidence provenance

The public source repository is github.com/yasmeenmh90-beep/flyrank-ml-internship-starter, with the research notebooks under work/notebooks/. The stages used here are w01_research_question, w02_ml_task_framing, w03_data_contract, w03_feature_leakage_check, w04_signal_audit, w04_baseline_score, w05_model, w06_validation_audit and w07_action_playbook. Reported software versions were NumPy 2.1.3, pandas 2.2.3 and scikit-learn 1.6.1, with seed 42 for the grouped modeling split.

Provenance statement. All numerical results and charts in this paper were transcribed or redrawn from the five uploaded ML-02–ML-10 research reports, themselves based on saved notebook outputs. They were not independently recomputed against FlyRank's restricted warehouse for this revision. A complete rerun requires authorized warehouse access, verified notebook paths and a fresh top-to-bottom execution. Exact repository availability of individual artifacts should be checked before public submission.

Acknowledgments and data credit

Built on the FlyRank ML Internship dataset. Dataset ownership and internship context are credited to FlyRank. Research synthesis and interpretation: Yasmeen Azmat Ali. This public paper contains aggregate results only.

References and source artifacts

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